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Evolutionary Technology and Phenotype Plasticity: The FATINT System

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Summary. This is a comprehensive summary paper of works done by the authors in 2004-7 but never published as a single narrative. Our works address various aspects of an Evolutionary Technology. We present the challenge, the approach (using simulated interactions from phenotype plasticity called 'fat interactions') and the first results. On the road, we develop and test a platform for the general agent-based modeling of biological populations, which serves as the evolutionary engine behind our FATINT system.

1 Introduction: Evolutionary Technology

The direct route to artifacts is via design, i.e. by specification and subsequent realization. Evolutionary Technology, as defined in [21], looks for an indirect route. The task is to produce an abundance of forms and functions of a practically endless variety by means of evolutionary methods. This implies the twin challenges of the 'arrow of complexity' and 'open-ended evolution', i.e. of producing increasingly complex machineries in the course of time, and doing that in a persistent, self-supporting process propelled by entirely endogenous causes.

Open-ended evolution is widely recognized as a difficult and unsolved problem. John Holland, the founder of Genetic Algorithms has recently asked: "Can we produce an existence-proof model, akin to von Neumann's model of self-reproduction, that exhibits open-ended evolution, with increasing diversity and complexity? " [16]. Also, the 'arrow of complexity' hypothesis is challenged from several directions ([2]), for a discussion see [22]. Currently there is no accepted general evolutionary theory for the origin of complexity or the maintenance of evolutionary change. Recent attempts to overcome one or both difficulties come from Artificial Life (such as the AVIDA system, [1][4]) and robotics such as the renowned GOLEM project [32] or swarm-based developments [27].

In the above terms, we understand Evolutionary Technology as an attempt to realize open-ended evolution showing complexification and to harness it for technological needs. The work reported in this paper deals with the first part of this problem (i.e. the growing space of functionalities and attributes), leaving issues of controllability aside.

2 Species, Niches, and Phenotypes

Our approach extends earlier work on evolving populations, such as Tierra, Avida, or Echo [35] [4] [17]. We maintain that in order to achieve open-ended evolution, the understanding of the structure and dynamics of evolving populations is crucial, and this is incompletely achieved with the current tools. This motivates our efforts to develop a comprehensive system with automatic observers for population modeling. Our open-ended evolution model is built on top of that.

Two fundamental concepts of biological populations that serve as the groundwork for our models are *species* and *niches*. To put it shortly, we attempt to build new species that recursively construct and fill new niches, giving rise to more species and more niches in the course of time. The third component of our approach is phenotype plasticity arising from interaction.

In the simplest definition, a species is a set of co-reproducing organisms reproductively isolated from others ('biological species' concept, Fig. 1, [23]). Speciation, the process by which species give rise to more species, is difficult to achieve in both natural and model populations. A particular mechanism considered sufficient is sexual selection, e.g., the dynamic change of mating preferences [10] [20]). The current theory about the origin of the giraffe ([37], although debated, [3]) assumes that it is male fighting (called sparring or 'necking') that was the key to the giraffe's long neck and to the separation of a species, the pre-okapi, into the modern giraffe and the okapi. Valid or not in the end, the giraffe story feeds on the power of sexual selection, and is closely related to the idea of phenotype plasticity as we use it below.

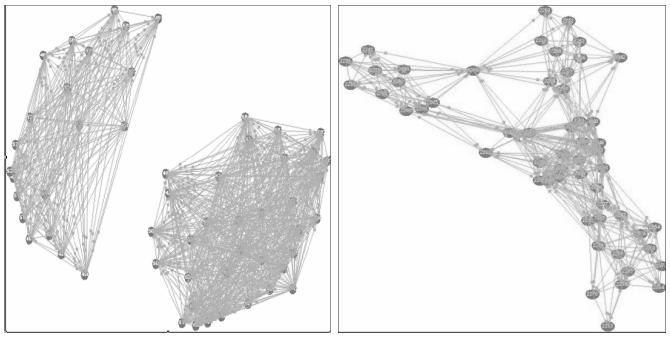

Fig. 1. Example species in a topological (Fruchterman-Reingold) plot, left; a realistic species from a simulation, right. Nodes denote individuals, edges represent pairwise reproduction ability. Species are identified as connected components

Niches are segments of a (multi-dimensional) resource space that can be occupied by a species ([18] [14]). Niche construction is a recently advocated theory of evolution that builds on mutual organism-environment interactions ([33] [29]) [12]). In niche construction, the central notion is the organism' activity that alters the (biotic or abiotic) environment in such a way which in turn can induce changes in the given (or a different) organism. A simple example is 'ecological inheritance': a new generation of organisms inherits not just genes but also a specifically altered environmental pressure left behind by the earlier generations.

Known theoretical models for niche construction tend to study consequences of organism-environment feedback on genetic composition (i.e. they are 'replicator level' models, in Dawkins' [6]terms). Laland et al. [28] considers a fixed genome two locus model where one locus operates controls on environmental states and the other suffers consequences via environmental selection. Suzuki and Arita [38] use a modified Kauffman NKES model [26] to analyze alternative strategies of genetic adaptation and the induction of environmental change. As was first pointed out, however, by Lewontin [30][31] and later others (e.g. Dusek [11]), a niche construction process is the product of an underlying *physical* interaction between the organism and its environment. Lewontin's own example is a bug that lives on the upper half of a leaf. He discusses how the bug can alter the evolutionary environment for its descendants by a variety of selected actions, as simple as moving to the other side of the leaf. A flexible phenotype expression (that adheres to a reaction norm) has the upper hand in such a process. This places the *interactor*, i.e. the phenotype, into the focus of interest.

The third source, phenotype plasticity, is one of the oldest ideas in evolution [25]. Yet until recently, little attention was paid to it. Phenotype plasticity can take various forms that includes, but is not limited to, function change ([5] [36]), exaptation [13], or 'tinkering' (*bricolage*, [19]). In our works the concept is used in a specific sense, for which the giraffe example offers a case. Here, unused traits (such as the length of a neck) become significant for interaction. This bears upon the very phenotype definition. Narrowly taken, the phenotype denotes those organismic properties that are relevant for an ecological-evolutionary context at a given moment [9]. We will consider organisms with changing phenotype definition. We treat phenotypes as variable entities we call *fat phenotypes* [22]. A fat phenotype is essentially a multiple phenotype that permits different, ecologically defined interactions. We note that every natural phenotype is 'fat' in the sense that the number of potential phenotype traits is usually very large compared to the focal/functional ('active') phenotype properties at a moment (Fig. 2)

Linking together Evolutionary Technology (i.e. open-ended evolution) with the species concept, niche construction and plastic phenotypes, we advanced a general Phenotype Hypothesis, PH, [22]. It predicts that no evolution model and no evolutionary technology/ALife design will be able to produce a sustained evolutionary process or support the increase of functional complexity without admitting a changing phenotype space as a natural part of the interaction potential of the organism.

In this paper we deal with testable consequences of PH. We will consider emergent species that occupy dynamically created niches based on phenotype interaction and examine two Hypotheses.

Hypothesis1. An evolutionary engine with phenotype-based emergent species and without phenotype plasticity remains stable and non-productive of species.

Hypothesis2. The introduction of phenotype changes in the above systems leads to a sustained production of new emergent species.

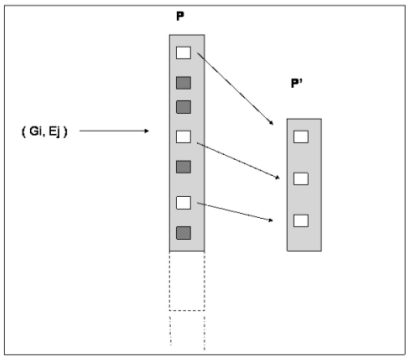

Fig. 2. Fat phenotype (P). A focal/functional phenotype P' is constructed by genes (G_i) and environment (E_j) together

3 The FATINT System

To address the above crucial issues of phenotype representation and to study their effects in maintaining open ended progress, we perform an extended study using digital organisms. We are developing and using the FATINT system (the name derives from Fat Interaction and hence from fat phenotype), which is part of the EvoTech project run at Collegium Budapest and Eötvös University. We reported early work on this system in [20] [22] [23] and elsewhere. The EvoTech home page is found at *www.evotech.hu*, with downloadables, papers, and technical reports.

The FATINT system was written in a consistent fully encapsulated agent-based approach [15], using the Java based REPAST environment developed by the University of Chicago [39]. The FATINT system can also be compiled into a stand-alone application. It offers a graphical user interface (Fig. 3) allowing for a wide selection of models with a broad range of interactive experiments using different parameter settings. The system is equipped with a high number of automatic (runtime) observers that include a metric distance plot, various histograms, population plots (such as number and age distributions, and programmable 2D projections), species graphs (using Fruchterman-Reingold and Kamada-Kawaii graph algorithms) and others.

For simplicity we fix the representation of the domain as follows. Each agent has its phenotype represented as a vector v_i of integers from the interval $[V_{min}, V_{max}]$. The length of the vector is L and is always identical for all agents but variable in time. Formally, the phenotype of agent i is:

$$v_i \in [V_{min}, V_{max}]^L$$

Fat phenotypes are represented in various ways as associated with either growing dimensions (i.e. variable length vectors) or as dimension switches (using silent and active values of vector entries). We experimented with both solutions (and some others [22]).

To accommodate phenotypic plasticity, note that the dynamic notion of phenotype applied here is global. Our phenotype definition is global in the sense that, if a given phenotype trait in organism A enters a particular interaction, then the

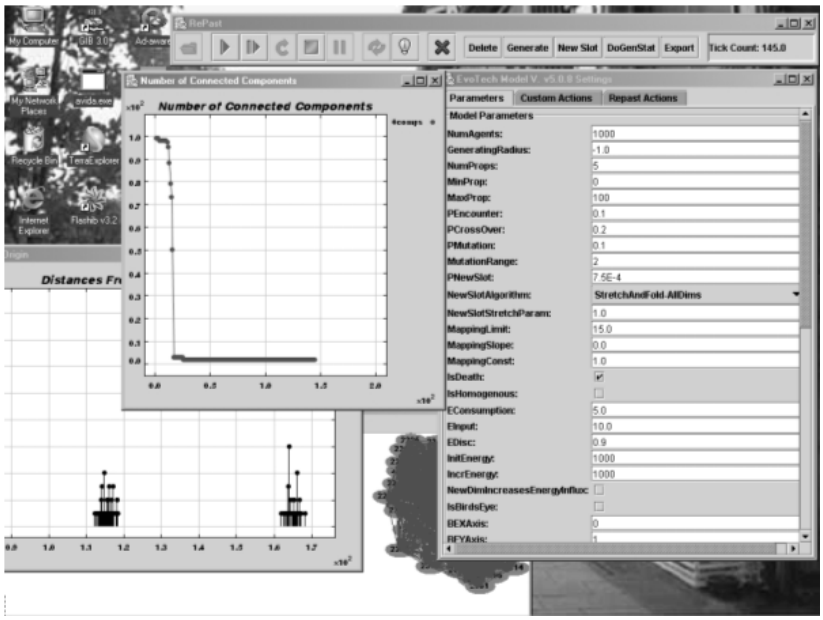

Fig. 3. Snapshot of FATINT's graphical user interface.

same trait typically also appears for organism B. (If one giraffe fights with a neck, others should defend themselves using theirs.) The notion is also dynamic in that it permits the change of the phenotype during the lifetime of an organism.

Table 1 presents different phenotype transformations [20], their causes (internal or external) and extension (local or global), corresponding to different types of genotype-phenotype maps (GPMs). In the current version of the system, we tested two families of GMPs: a type-dependent and a type-independent one [20] [22]. In the first, phenotype change occurs in an identical way for all organisms that share a given genotype, whereas in the second, this is not the case (the two situations approximately correspond to the first and the last rows of Table 1. We return to the issue in Section 6.

Experiments were conducted using a basic evolution engine, the task of which was to maintain a stable population before developing new species, as well as its extension with a phenotype change support, which is an independent system component.

4 The Evolution Engine

In FATINT we currently use sexually reproducing gender-less (i.e. 'snail-like') agents, which represent individuals. Unless stated differently, genotypes and phenotypes are always assumed to coincide; however, the model's behavior is based on the phenotype interpretation. Reproductive success depends on a phenotype similarity metric which introduces a sexual selection. The resulting selection force is variable

Table 1. Mechanisms of phenotype change

Form	Cause	Type
Point mutation	endogenous	local
Phenocopies	exogenous	partly global
Epigenetic change	both	partly global
Horizontal adaptation	both	global
Behavior change	exogenous	global

(any individual can become the 'center' of a species) yet well-defined (i.e. once a center is formed, 'distant' individuals are selected against). In the FATINT system, a species emerges as a dynamically maintained cluster of interbreeding individuals, around a dynamically defined center characterized by the predominant types.

At the base level the system uses an evolution engine that consists of a population of agents using a single non-replicating resource (energy), as well as usual evolutionary operators for crossing over, mutation, resource uptake and ageing/death. In addition to having a phenotype, agents only have the minimum of properties: age and accumulated energy.

Each organism has an equal chance to 'eat' at every time-step. Note that this nevertheless introduces an implicit competition for energy, which leads to density-dependent effects (genotypes in higher numbers obtain a larger share of energy). A consequence of the above implicit competition is competitive exclusion [23] [8], that is, in the long run only one single species can survive. This puts an important control on our open-ended experiment in the present form and serves as a kind of negative feedback.

The evolution engine uses a quasi-parallel activation regime. Each agent acts once per every time step in a dynamically randomized order. The agents' activity consists of the steps of energy intake, energy consumption, and reproduction. The agent first seeks E_{in} units of energy in the shared environment, and, depending on the amount available, it receives e_{in} units (possibly 0). The efficiency of energy intake decays with age:

$$e_{accumulated} = e_{accumulated} + e_{in} \times (E_{discounting})^{age}$$

where $0 < E_{discounting} < 1$. Next, the agent consumes $E_{consumption}$ units of its accumulated energy. If it does not have this amount available, the agent dies. Surviving agents attempt to reproduce with probability $P_{encounter}$ (i.e. every time step produces part of an entire new generation). Updating the shared environment completes the iteration. This means the addition of $E_{increase}$ units to the energy pool.

In reproduction an active agent picks a random mate from the list of potential partners, which are individuals similar enough to bear an offspring with the given agent. Similarity is measured by the Euclidean distance between the agents' phenotype vectors. An advantage of this metric is that it is dimension independent. It allows for sexual selection to occur in the same way between any selected phenotype pairs of arbitrary dimensionality. (This choice has also known side effects. To test them, in [24] we discussed three different families of similarity operators, but no fundamental difference in speciation experiments were found).

Given two parents, and similarity d defined as above, the number of offsprings is $M_{const} + (M_{limit} - d)M_{slope}$ (for $d > M_{limit}$). New agents inherit the active parents phenotype, except for mutation and crossing over that occur with probabilities $P_{mutation}$ and $P_{crossing}$, respectively, per gene. Mutation shifts the value of a gene by a random value in $[-V_{mutation}, +V_{mutation}]$. If the mutated value falls outside the interval $[V_{min}, V_{max}]$, the offspring is dropped.

Table 2. Default parameter settings

$V_{min} = 0$	$M_{limit} = 15$
$V_{max} = 100$	$M_{slope} = 0$
$P_{encounter} = 0.1$	$M_{const} = 1$
$P_{crossing} = 0.2$	$E_{consumption} = 5$
$P_{mutation} = 0.1$	$E_{in} = 10$
$V_{mutation} = 2$	$E_{discounting} = 0.9$
	$E_{increase} = 1000$

5 Behavior of the Basic Evolution Engine

First we ask what happens if we initialize the basic system with a randomly chosen set of initial conditions and let it run with the default parameters or different parameter settings. Our results are summarized in Fig. 4.

On the Figure we see that the evolution engine, left alone, converges to a single stable species and maintains that indefinitely under a very wide range of parameters and initial conditions [20]. Initial conditions were defined as randomized populations, i.e. where each individual had a randomly defined genotype and phenotype. Fig. 4 (above left) summarizes 10 different runs with different random seeds. The solid line show the average of ten runs, grey indicates the envelope of areas covered by any run. It is seen that the convergence speed is a function of the initial conditions but otherwise convergence and the subsequent maintenance of a single species is unequivocal. From other representations, e.g. distance plots or histograms (not shown here) we can conclude that each random initial condition yields a different surviving species with a different emergent center in phenotype space. This typical behavior shows the development and stability of biological species based on phenotype-to-phenotype interaction alone.

After demonstrating this typical behavior we want to test its limits. To that end first we varied encounter frequency, a parameter with a strong effect on reproduction and hence long-term survival (Fig. 4, upper right). Each solid line shows an average of 10 runs with different initial conditions. As expected, low values of $P_{encounter}$ make reproduction difficult or impossible; for values higher than 0.07 the typical behavior is found; no speciation events occurred at any value. Varying $P_{mutation}$ (Fig. 4, lower left) had little effect. This counter-intuitive outcome is due to the fact that speciation in the model population is not based on external fitness values but relational (i.e. pairwise) reproduction events among organisms. Such events are not

easily facilitated by single distant mutants, and as the next experiment shows, news species must typically emerge from within the hyper-sphere containing the currently predominant or 'old' species. Fig. 4 (lower right) shows the effect of varying crossing over. At high frequencies of crossing over, a high internal variability is produced within the diameter of the existing species, and this leads to several new pairwise matches among reproduction candidates, resulting in a temporary instability of the surviving species.

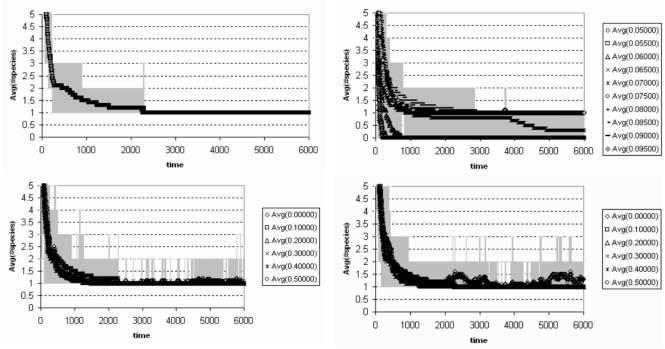

Fig. 4. Number of connected components (vertical axis) against time. From top-left: basic engine behavior with default parameters; effects of varying $P_{encounter}$(in the range $0.05 - 0.095$), $P_{mutation}$ (0.10.5), and $P_{crossing}$ (0.10.8). 10 different random seeds; $6,000$ steps

To answer questions about the 'reality' of species found by the method of connected components above, in a study involving three different automatic species observers [23] we demonstrated the robustness and realistic behavior of the evolution engine. Fig. 5 (left) shows convergence to a single cluster in phenotype space ('phenetic species'), using a standard hierarchical clustering method, characterized with a clustering constant $d_{cluster}$. We express $d_{cluster}$ here as a ratio to M_{limit}: at $d_{cluster} = 1$ we ask about clusters of size M_{limit}.

Fig. 5 (left) demonstrates that even at small values (e.g. $d_{cluster} = 0.9$) the same stable convergent behavior as in Fig. 4 is found. In intuitive terms, this means that the 'diameter' of the species in n-space is about as small as M_{limit} in the long run or smaller (or otherwise, smaller values of $d_{cluster}$ should split the population into different phenetic species). Using a cladistic species concept [23] that asks for lineages we obtain a similar picture (Fig. 5, right): all individuals in the long run stem from one single lineage. A similar coincidence of the various species concepts will be demonstrated (by calculating the Rand index [34] of various classifications) during the speciation experiments, except at the time of the very speciation events.

The FATINT evolution engine also permits further studies such as drift, Hardy-Weinberg equilibrium, density-dependent effects and other phenomena which are well understood at population level (i.e. mean field) models but require further study in the present individual-based framework (which we began in [8] [7]).

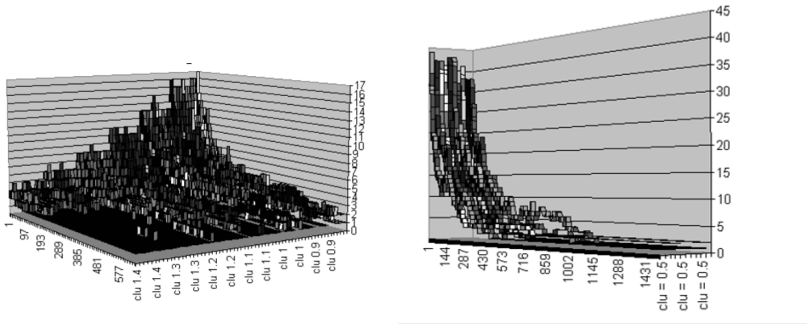

Fig. 5. Convergence. Left: clusters against time. Summary of 60 runs: 10 for 6 random seeds each, from $d_{cluster} = 0.9$ to 1.4. Time goes left to right, $d_{cluster}$ increases right to left. Vertical axis shows number of clusters. Right: cladistic analysis of the same 6 populations. Clades up to a given time. Time as before, vertical axis shows clade numbers

6 New Species and Functions from Phenotype Change

To test the effects of the phenotype changes, in FATINT's phenotype change mode, at every time step a phenotype change can be initiated by hand or, alternatively, at every reproduction event a new phenotype dimension can be introduced with a variable probability P_{change} per offspring. When such an event occurs, a new component is added to the agents' phenotype vector. The particular phenotype value an agent receives depends on the used method (we call 'stretch' method: the name refers to the degree new phenotype values are stretched along the new dimension, a factor that will prove to be important).

As alluded to in Section 4, two basic methods were applied (genotype-based, and genotype independent). The type-based stretch method calculates the value from the agent's old geno/phenotype. For simplicity, only the value v of a single dimension (the last one) is used:

$$v_{new} = V_{min} + (v \times V_{stretch}) mod (V_{max} - V_{min} + 1)$$

where $V_{stretch}$ is a positive parameter. The type-independent method selects a uniform random value from $[V_{min}, V_{max}]$.

To understand the effect of phenotype changes discussed in Section 3, we first study the hand-introduction of a phenotype change. The results for a typical run are shown in Fig. 6.

An initially compact and homogeneous species becomes structured and, in the course of time, this structure is amplified to give rise to two new species. Structuring occurs in terms of reproductive links formed and broken: i.e. the effect of phenotype change is to introduce new differences and also new similarities among organisms. Due to repeated success and failure of reproduction events both of them enter a positive feedback loop (Fig. 6, middle panels) which results in an amplification of the changes. Note, however, that the basic engine behavior as analyzed before acts against this phenomenon, and so the result can depend on several, usually random factors. Hence, speciation is bound to be a contingency-driven effect for

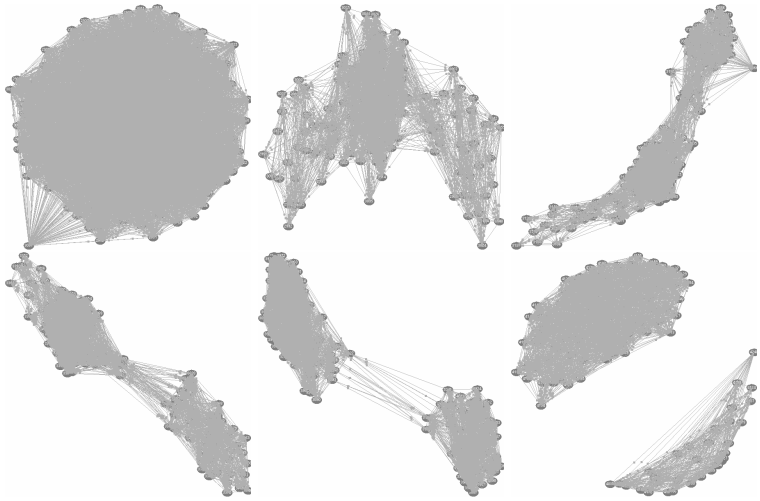

Fig. 6. Phenotype plasticity induced speciation in a sample run. New phenotype is introduced at left. First a homogenous reproduction network becomes structured, which is further amplified. After several generations, the process leads to segregation and the emergence of new reproduction centers. Positions are illusory in this topology plot.

which phenotype change appears to be a necessary, but itself not sufficient, condition in our model.

We experimented both with the type-dependent and the type-independent method (Fig. 7, right panel). We found in each case an increased heterogeneity of the system along the new phenotype dimension (see Fig. 7 bottom left for the type-dependent case). The new dimension is represented in the rightmost histogram. In line with our analysis of the stability of the basic engine in Section 5 we see several new candidate species centers arising (Fig. 7 top left), most of them *within* the diameter (usually: within the M_{limit}) of the original species. Later dampening and amplification events tend to sort out the majority of these candidates, or all of them, leaving one (the original) or two species (the old and a new one) as survivors. We observed rare 3-speciation events as well. The new species are characterized by new species centers and hence different functionalities (i.e. different sexual selection preferences).

We want to understand the process of phenotype change and the speciation transition more closely. Species analysis tools presented in [23] as well as the histograms (Fig. 7) provide a cue. Using either method, new phenotype traits turn out to be highly variable. For example, a sample run produced the comparisons seen in Table 3. Stable species situations before speciation are characterized by a Rand index $R = 1$ for all pairwise comparisons of the species metrics ([23]). Around the time of the introduction of a new phenotype trait ($t = 450$), clusters formed using phenetic (similarity-based) and cladistic (lineage-based) classifications strongly diverge from each other (whereas the former correlate well with the clusters of reproduction ability under a biological species notion), showing that phenotype properties are

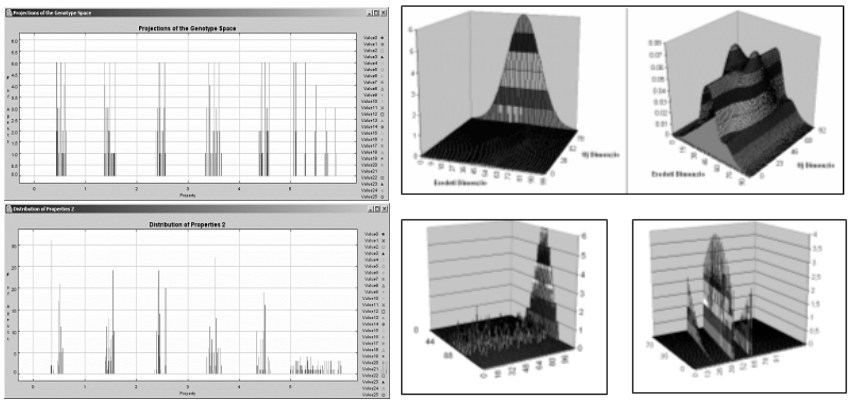

Fig. 7. Left panel: allele histograms at the time (bottom) and after a phenotype-driven speciation event (top). Histograms show the population distribution of alleles for a given gene. For modeling reasons, the picture shows a 5 gene 100 allele situation and its transition to 6 genes or phenotype traits. Right panel: cartoon fitness functions(top) and empirical distributions (bottom) before and after phenotype change for the type-independent (bottom left) and type-dependent (bottom right) model

more different than are lineages: the same lineage supports a number of different phenotypes. The situation changes after species segregation has taken place (around $t = 550$). By the time the two new species are completely formed ($t = 1000$) the same conditions as before speciation ($R = 1$) are restored.

This high variability of new phenotype traits at the time of transition is a simple mathematical consequence in the model and a natural assumption about real populations, where unused properties do not experience stabilizing selection. As a consequence, the population diameter increases along these dimensions and thus when a new dimension is activated new close matches (new types of reproduction events) become possible.

To summarize: the typical process upon the introduction of a new phenotype trait is the appearance and amplification of phenotype heterogeneity which often leads to the appearance of new emergent species centers 'strong' enough to facilitate the separation of the formerly stable species into new species showing different functionalities (in the limited sense of different sexual selection forces).

Table 3. R index of biological (BSC), phenetic (PSC) and cladistic (CSC) species at the time of a speciation event starting at $t = 450$

Time interval	PSC vs. BSC	PSC vs. CSC	BSC vs. CSC
450-550	R = 0.716	R = 0.356	R = 0.640
515-550	R = 0.716	R = 0.716	R = 1.0
550-1000	R = 1.0	R = 1.0	R = 1.0

7 Autonomous Speciation Experiments

In the autonomous phenotype change mode, as noted above, at every reproduction event a new phenotype dimension is introduced with probability P_{change} per offspring. To study the effect of this behavior we introduce nonzero values of P_{change} (Table 2). The findings are summarized in Figure 8.

The Figure shows that the autonomous mode of FATINT typically produces sustained speciation events (see Figure 8, left, and [23]). We find repeated speciation and subsequent relaxation. A developing species can be stable for hundreds of generations but tend to collapse afterwards due the factors mentioned in Section 5. If, before a complete relaxation repeated phenotype transitions occur, this can drive the system into a sustained multi-species state (Fig. 8 left) or into a runaway condition (Fig. 8 bottom right). To the interpretation of the Figure, we note that the typical population size using the default parameters was around a few hundred individuals (the batch experiments are computation intensive and were performed on a Beowulf cluster taking typically a week). Population size puts a serious limitation to speciation: a long term survival of 10 species means that an average of a few dozen individuals are allocated per species. Turning things around, we get that no more than 10 species can meaningfully exist for a longer time. Another important limiting factor is deliberately introduced density dependence (Section 3) which puts an important control by making the system conservative (i.e. dominated by negative feedback). This results in a short lifetime of the generated new species and forms a condition that can be dropped by using different resource models in the future ([8]).

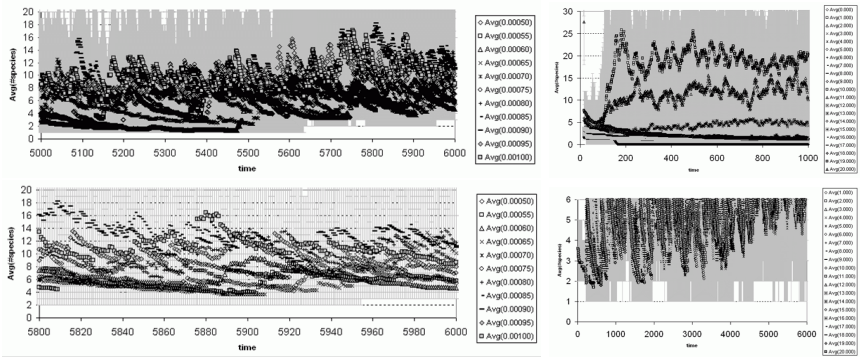

Fig. 8. Left panel: sustained speciation under various values of P_{change}. Right panel: top, decreased levels of M_{limit} $(5-15)$ disrupt species. Bottom, increasing phenotype modification $(V_{stretch}, 1-20)$ leads to species proliferation. Black line: average, grey lines, error bars, 10 runs. Time goes to right, species numbers are on vertical axis.

Figure 8, left, delivers the main results of our simulations. It shows that phenotype plasticity can drive an otherwise stable system into speciation repeatedly and push species numbers to high limits marked by population size and other factors.

We also wanted to study the behavior of the autonomous system under various parameters. Two important parameter choices are presented in Figure 8, right panel, otherwise using default parameters. As before, solid lines represent averages over 10

runs, grey cover the area spanned by all runs. (We omitted here the studies of mutation and crossing over which produce similar results as seen before; this and other items of sensitivity analysis are found in [20]). At the top, we see experiments with varying M_{limit}. As expected, this is a sensitive parameter and its low values can artificially disrupt species, a method with a further potential to exploit in future modeling studies (e.g. evolving M_{limit} in the course of the simulation). Increasing M_{limit} alone does, however, not form species having well-defined new properties (and consequently, stable high values of R). The other parameter tested is $V_{stretch}$ which is responsible for the variability of new phenotype traits. Clearly, low (or zero) values decrease or cancel any effect from phenotype change, whereas high values lead to a dramatic increase of variability. Correspondingly, we find runaway conditions in speciation for high values of $V_{stretch}$ (Figure 8, bottom right).

8 Conclusions and Future Works

We presented the development and results of an interactions-based evolutionary model (in progress for more than 3 years). The model uses similarity-based sexual selection to approach open ended evolution.

Answering *Hypothesis1*, we have shown that a phenotype based system shows species convergence and stability under the specified conditions. To test *Hypothesis2* we demonstrated that changing phenotype-to-phenotype interactions can repeatedly split species by the production of new selection constraints. We hypothesized that sustained ecological evolution always proceeds by similar mechanisms. In FATINT, the presented speciation process can produce new functions (i.e. new sexual selection processes) and can go on indefinitely using different mechanisms for phenotype change as genetic point mutations (our type-dependent method) or behavior change (type-independent method).

Topics that go beyond the limits of this paper include competition avoidance [22], the analysis of the role of positive and negative feedback in open-ended evolution [23], and our first results on emergent ecosystems using various genetic models and genotype-phenotype maps [8][7].

Finally we ask about the relationship to the distant goal of evolutionary technology. The presented works clearly signify first steps in this direction, addressing what appears to be a main bottleneck, namely the understanding of the role of material structures (i.e. 'embodiment'). Our results imply that *fat phenotypes* may play a key role in pushing speciation forward under specific conditions. A next step would be to extend these results to functional differentiation (e.g. ecosystems) and to the study of body plans (e.g. using evolvable genotype-phenotype maps).

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